GOGGLES

Geometry-Optimized Global Grounding for Localization and Ego-motion Sensing

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State Estimation

Someone asked earlier this quarter

“so this is a solved problem, right?”

We find that despite all the math and covariance matrices we’ve thrown at it, state estimation is still a tough problem.

OpenVins

Classical solution for Visual Odemetry

- **Visual Inertial Odometry**
- FAST **feature detector**
- Kanade-Lucas-Tomasi **Optical Flow**
- MSCKF state estimation and prediction
- SLAM landmarks for loop closure
- Real-time (100Hz)

Domain: Pose Estimation via Passive Perception

Accurate Pose Estimation

Difficult without

1) **external pose estimation (GPS, MoCap)**
2) **OR expensive active perception (LiDAR)**
- Cost, high size, weight, power requirements

Passive Perception (Camera-only)

Remains paragon of State Estimation

!! But modern foundation models claim to produce accurate pose estimates through massive pre-training. !!

Explore: Classical methods vs. Foundation models

CLASSICAL

Visual Odometry

- Brittle in unstructured environments
- Relies on sparse feature tracking

MODERN

Geometric Foundation models

- Achieve robust pose estimation
- But too computationally intensive for real-time deployment on robot hardware

Despite their claims of geometry understanding, modern methods still require filtering to capture full odometry from a scene.

Problem: Benchmark methods, Evaluate Filtering

Simulated Trajectories ✅ → **Benchmark Methods** ✅ → **Evaluate effect of Filtering** #TODO

1. Capture 3 scenes
2. Train Gaussian Splats
3. Create 12 custom trajectories

1. Classical Methods
   - OpenVINS
2. Modern methods
   - Reloc3r
   - StreamVGGT
   - Depth-Anything-3

1. Implement Filtering
2. Evaluate effect on each model

Approach

Simulate closed-loop quadcopter dynamics & estimate relative camera pose from images.

Evaluate Translational & Rotational Error

- Mean, Median, Area Under Curve

Benchmarks

OpenVins

- **Visual Inertial Odometry**
- FAST **feature detector**
- Kanade-Lucas-Tomasi **Optical Flow**

- MSCKF state estimation and prediction

- SLAM landmarks for loop closure

- Real-time (100Hz)

CLASSICAL

Foundation Models

Reloc3r (2024)
StreamVGGT (2025)
Depth Anything v3 (2025)

Feed-forward Vision Transformers
Trained on massive datasets (millions of images)
Predict:

1. Camera poses & intrinsics
2. Depth estimates
3. 3D pointclouds

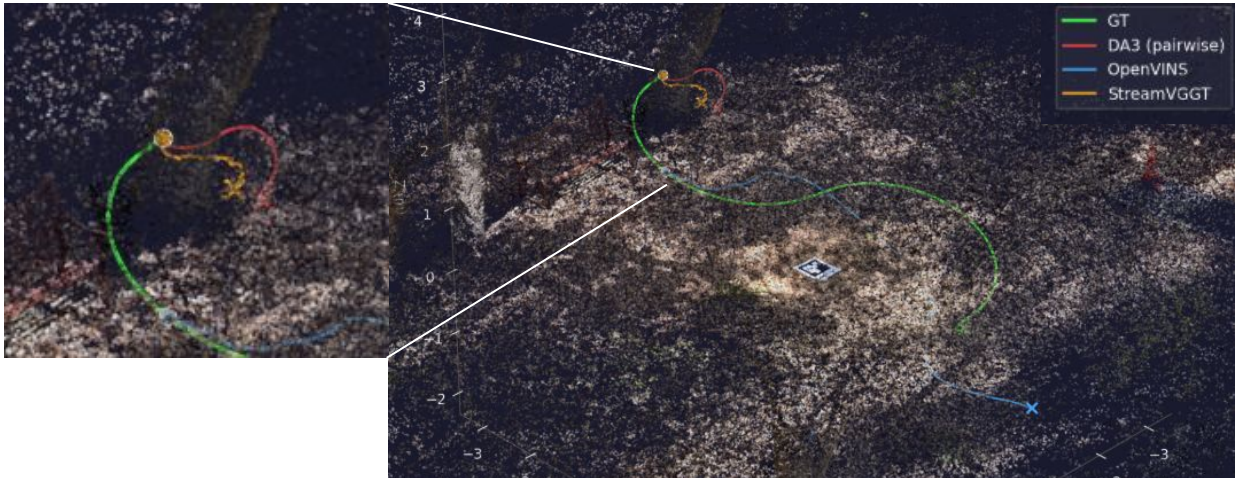
Slow (3-4Hz)

MODERN

Models

Model Summary (averaged across scenes/courses)

Model	N	Rot Med (°)	Rot Mean (°)	Trans Med (°)	Trans Mean (°)	AUC@5	AUC@15
openvins	36	44.09	46.91	43.21	44.41	0.00	0.03
streamvggt	36	**9.22**	12.55	15.67	19.51	**0.13**	**0.35**
reloc3r	36	41.88	51.13	41.22	42.89	0.03	0.08
da3_pairwise	36	9.41	**10.29**	**12.49**	**16.15**	0.03	0.24



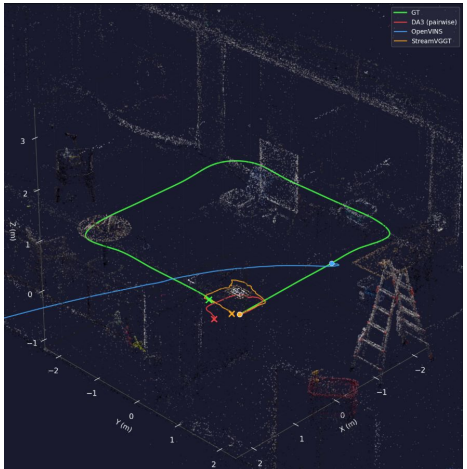
Courses

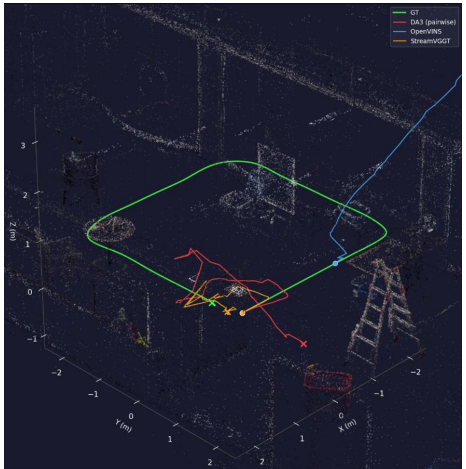
The camera orientation still governs pose estimation quality

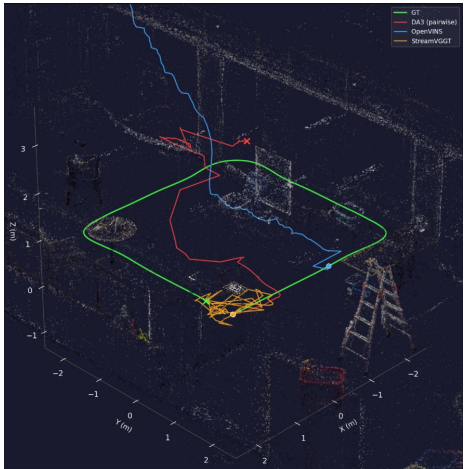
Outside-In

Inside-Out

Along Track





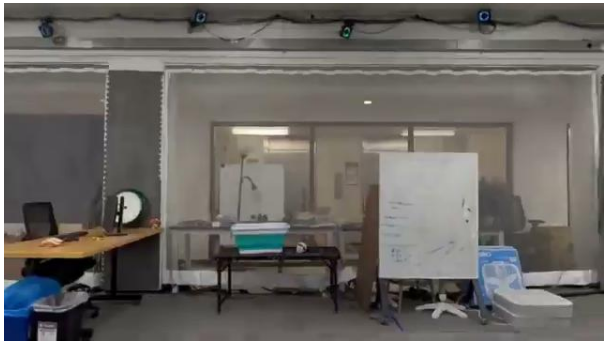


Scenes

Feature sparsity and scene regularity degrade model performance

Packardpark - dense

Flightroom - moderate



Button - sparse

Scene Summary (averaged across courses/models)

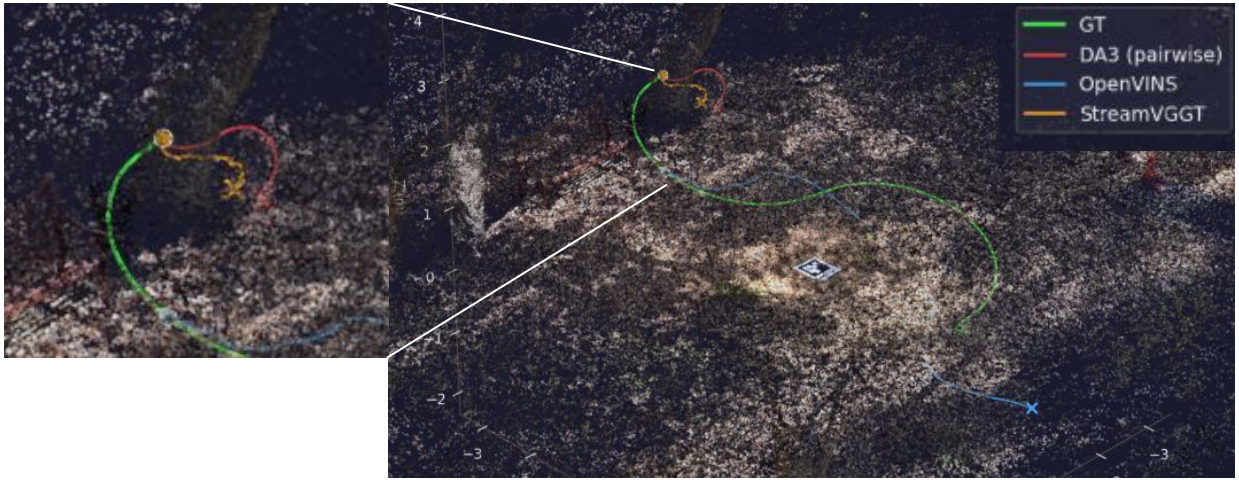
Scene	N	Rot Med (°)	Rot Mean (°)	Trans Med (°)	Trans Mean (°)	AUC@5	AUC@15
button_drywall_on	36	21.87	24.25	**23.21**	**25.60**	0.04	**0.21**
flightroom_ssv_exp	36	22.75	26.31	23.36	26.43	0.06	0.21
packardpark	36	**18.09**	**19.18**	24.80	28.04	**0.06**	0.20

Conclusion

Classical methods don’t work out of the box

Foundation models get us close (“metric” but not scale accurate)

This motivates a hybrid approach where IMU filtering is applied to camera pose estimates from foundation models



Q&A

Problem Statement

In the real world, accurate poses are difficult to obtain without external pose estimation (GPS, MoCap), or expensive (high size, weight, power requirements) active perception (LiDAR).

Pose estimation through passive perception (cameras) remains a paragon of state estimation.

Foundation models claim to produce accurate pose estimates through massive pre-training.

We explore the gap in state estimation between:

Classical visual odometry

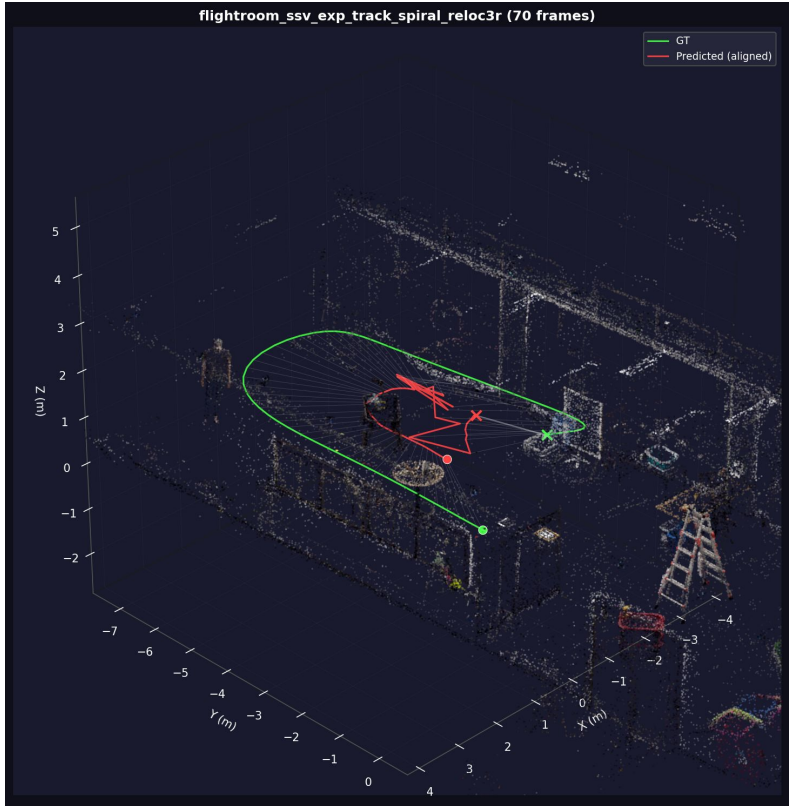
- Brittle in unstructured environments
- Relies on sparse feature tracking

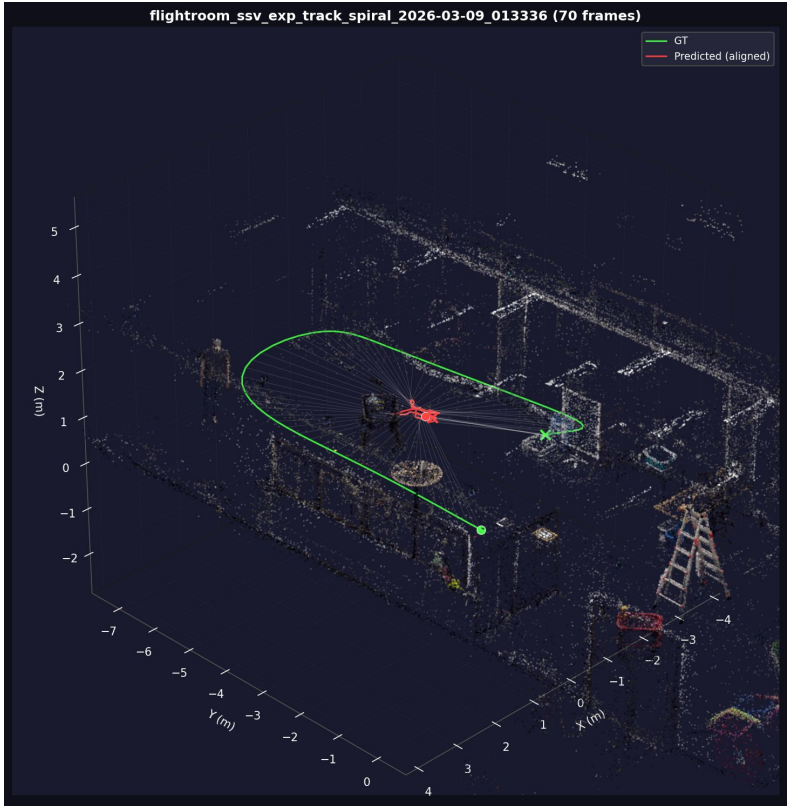
Modern geometric foundation models

- Achieve robust pose estimation
- But too computationally intensive for real-time deployment on robot hardware

We find that despite their claims of geometry understanding, modern methods still require filtering to capture full odometry from a scene.

Reloc3r vs StreamVGGT (track_spiral)

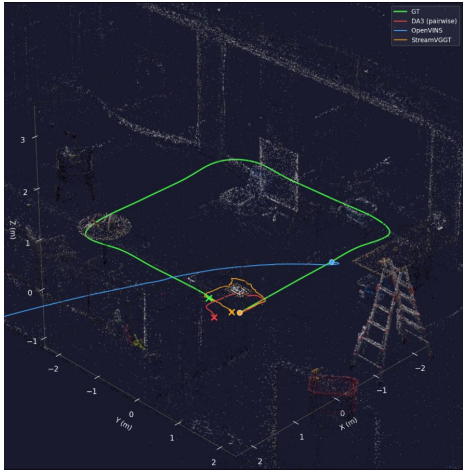




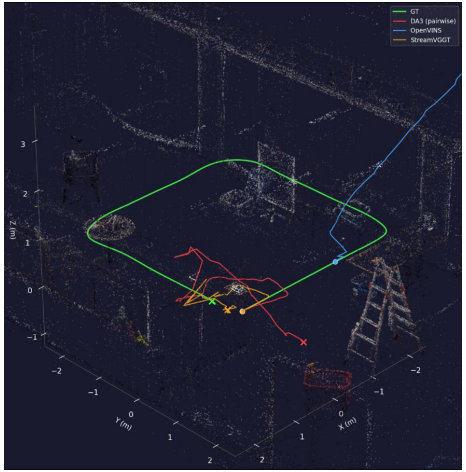
Courses

The camera orientation still governs pose estimation quality

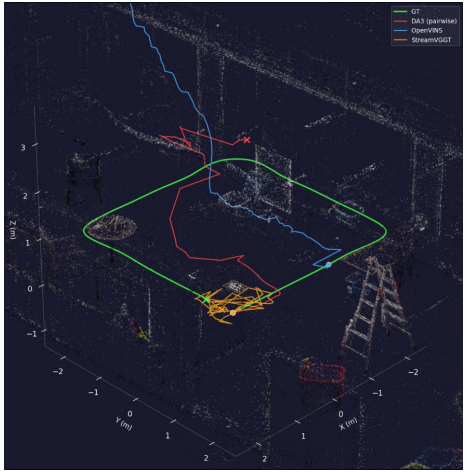
Outside-In



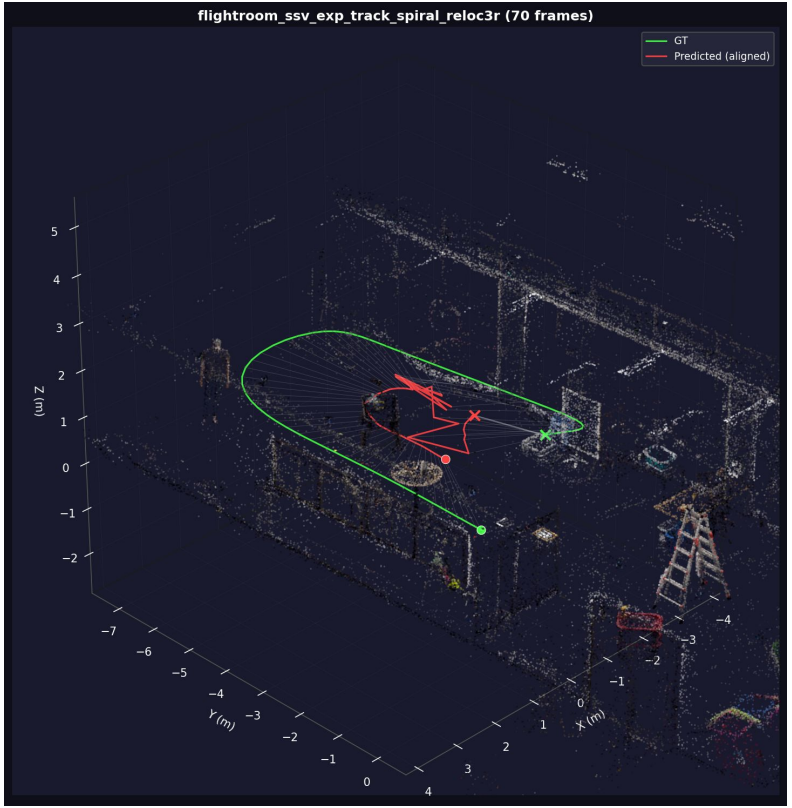
Inside-Out



Along Track



Reloc3r (track_diagonal)



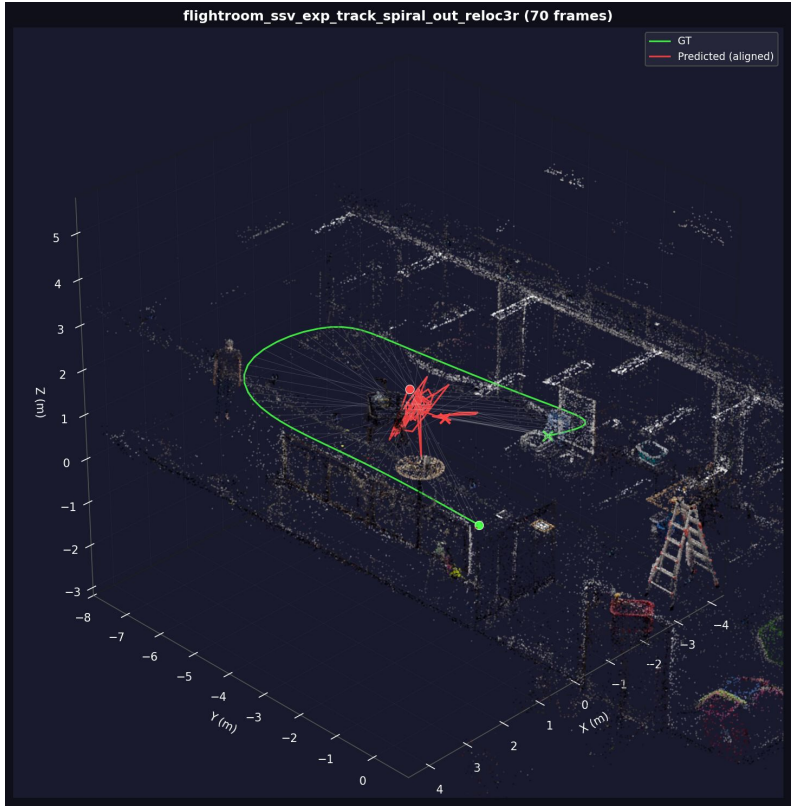

```
============================================================
Relative Pose Evaluation Results
============================================================
  Pairs evaluated: 2415

  AUC Metrics:
    AUC@3:  0.0257
    AUC@5:  0.0569
    AUC@15: 0.2819
    AUC@30: 0.4891

  Rotation Error:
    Mean:   2.454 deg
    Median: 2.077 deg
    Acc@5:  0.9317
    Acc@15: 1.0000

  Translation Error:
    Mean:   20.428 deg
    Median: 12.974 deg
    Acc@5:  0.1288
    Acc@15: 0.5545
============================================================
```

Reloc3r (track_spiral - outward facing)



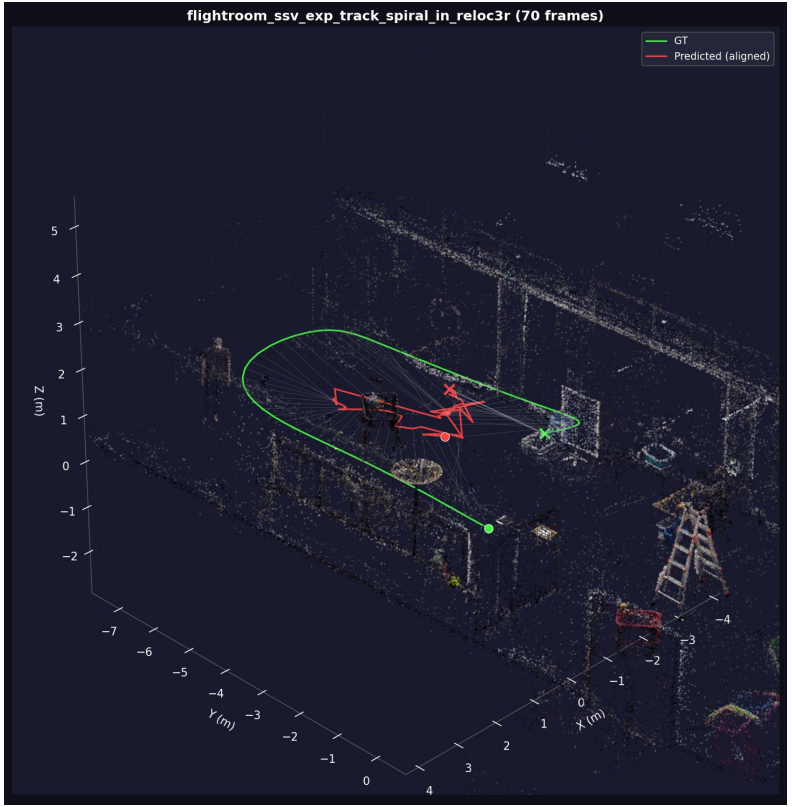

```
============================================================
Relative Pose Evaluation Results
============================================================
  Pairs evaluated: 2415

  AUC Metrics:
    AUC@3:  0.0003
    AUC@5:  0.0022
    AUC@15: 0.0193
    AUC@30: 0.0749

  Rotation Error:
    Mean:   53.999 deg
    Median: 20.278 deg
    Acc@5:  0.2170
    Acc@15: 0.4447

  Translation Error:
    Mean:   44.154 deg
    Median: 41.269 deg
    Acc@5:  0.0199
    Acc@15: 0.1168
============================================================
```

Reloc3r (track_spiral - inward facing)



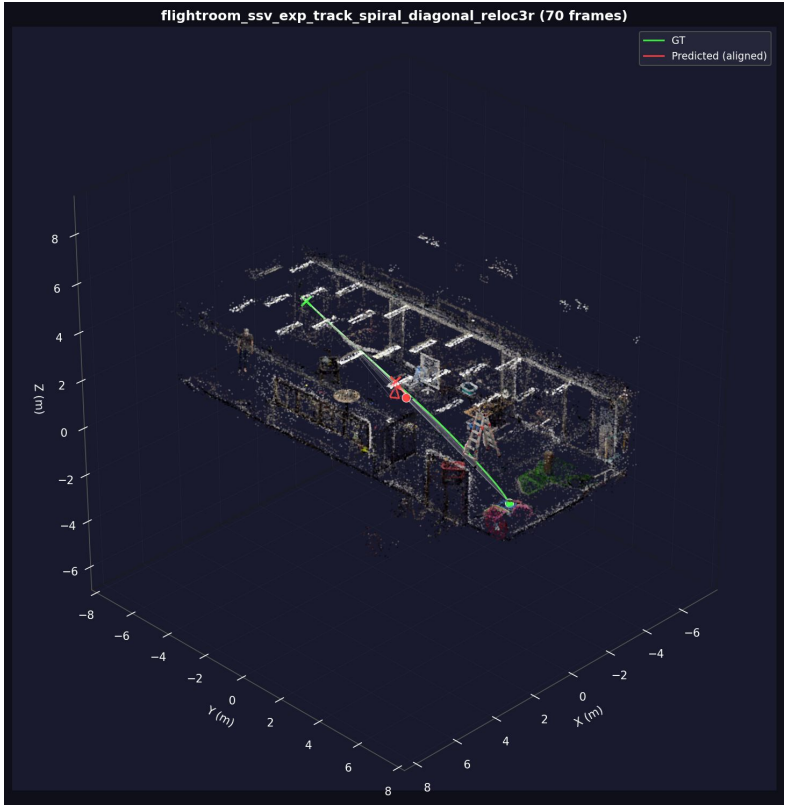

```
============================================================
Relative Pose Evaluation Results
============================================================
  Pairs evaluated: 2415

  AUC Metrics:
    AUC@3:  0.0025
    AUC@5:  0.0100
    AUC@15: 0.1026
    AUC@30: 0.2304

  Rotation Error:
    Mean:   3.287 deg
    Median: 2.627 deg
    Acc@5:  0.7781
    Acc@15: 1.0000

  Translation Error:
    Mean:   40.044 deg
    Median: 36.447 deg
    Acc@5:  0.0331
    Acc@15: 0.2605
============================================================
```

Reloc3r (track_spiral - diagonal)




```
============================================================
Relative Pose Evaluation Results
============================================================
  Pairs evaluated: 2415

  AUC Metrics:
    AUC@3:  0.0062
    AUC@5:  0.0296
    AUC@15: 0.2278
    AUC@30: 0.4144

  Rotation Error:
    Mean:   4.779 deg
    Median: 4.062 deg
    Acc@5:  0.6029
    Acc@15: 0.9963

  Translation Error:
    Mean:   24.671 deg
    Median: 15.826 deg
    Acc@5:  0.1035
    Acc@15: 0.4812
============================================================
```